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BOOK CHAPTER

Contraception and abortion



Close

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prehensive Gynecology, 13, 238-254.e2

Key points

- Approximately 45% of all pregnancies in the United States are unintended, and among women who experience unintended pregnancy, more than half are not using contraception. By age 45, at least half of U.S. women will experience an unintended pregnancy, and one in four will have had an abortion.
- Failure rates in the first year of contraceptive use are highest for coitus-related methods (e.g., withdrawal, periodic abstinence, condoms, barrier methods) followed by combined contraceptives (pill, patch, ring) and the progestin injection. Intrauterine devices (IUDs), implants, and sterilization have typical use failure rates of less than 1%, similar to that of sterilization.
- Combined hormonal contraceptives increase a woman's risk of venous thromboembolism (VTE) about threefold to approximately 1 in 1000 per year. Women with risk factors for VTE or cardiovascular disease (e.g., obesity, age older than 35, smoking, a personal or family history of clotting disorder) should use effective birth control methods without estrogen.
- The most effective method of emergency contraception is the copper IUD, followed by a single dose of oral ulipristal acetate. An oral dose of levonorgestrel is somewhat less effective but available over the counter.

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infertile, and not trying for pregnancy. Of those at risk, about 10% do not use contraception. **About 45% of pregnancies in the United States are unintended, and among women who experience unintended pregnancy, more than half are not using contraception** (Mosher, 2010).

Tubal sterilization (18.6%), the oral contraceptive pill (12.6%), long-acting reversible contraceptive (LARC) methods (10.3%), and condoms (8.7%) comprise the most commonly used methods (Table 13.1)(Daniels, 2018). Distinct advantages and disadvantages characterize contraceptive methods. Clinicians should be able to explain the unique features of each method, must identify medical contraindications to a given method and offer safe and effective alternatives, and should keep the risks of pregnancy in mind when discussing methods. **For healthy women, and especially for medically challenging patients, pregnancy risks generally exceed risks related to contraception.**

 **TABLE 13.1**

Percentage Distribution of Women Aged 15 to 44, by Current Contraceptive Status: United States, 2015-2017

Data from Daniels K, Abma J. Current contraceptive status among women aged 15-49: United States, 2015-2017. *NCHS Data Brief*. 2018;(327). Available at <https://www.cdc.gov/nchs/data/databriefs/db327-h.pdf> (<https://www-cdc-gov.ezproxy.med.cornell.edu/nchs/data/databriefs/db327-h.pdf>) .

Characteristic	%
Using contraception	64.9
Female sterilization	18.6
Male sterilization	5.9
Pill	12.6
Male condom	8.7
Long-acting reversible contraceptives	10.3

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Contraceptive counseling

Health care outcomes improve when provider counseling incorporates patient preferences. In contraception counseling, communication between the patient and provider can affect initiation and continuation of birth control. Most women value personal autonomy when making decisions about contraception more than they do in other areas of medicine (Dehlendorf, 2010).

Shared decision making (SDM) provides a useful counseling tool. Using this model, the clinician contributes medical expertise, and the patient contributes her values and preferences. Priorities vary; therefore counseling should be tailored to each individual woman. SDM works particularly well in contraception counseling because most methods of birth control are safe for most women. **A shared decision-making approach correlates with improved patient satisfaction compared with decisions driven solely by the provider or the patient** (Dehlendorf, 2017).

Medical eligibility criteria for contraceptive use

Most women initiating contraception are healthy, and in general the risks of pregnancy outweigh the risks of most contraceptive methods. However, there are certain medical conditions that may contraindicate using a particular method. The health care provider must be aware of these specific contraindications to ensure that a given method is both acceptable to the patient *and* safe.

The **World Health Organization** (WHO) publishes detailed guidelines listing the medical eligibility criteria for the use of individual contraceptive methods. These regularly updated guidelines can be downloaded from the WHO website (www.who.int/reproductivehealth) (WHO, 2015). The **Centers for Disease Control and Prevention** (CDC) also publishes regularly updated medical eligibility criteria tailored to U.S. practice; CDC

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general, coitus-related methods and more user-dependent methods are less effective than “forgettable methods” such as LARC. **Combining any method with a condom decreases the risk of sexually transmitted infection acquisition.**


TABLE 13.2

Percentage of U.S. Women Experiencing an Unintended Pregnancy During the First Year of Typical Use and the First Year of Perfect Use of Contraception and the Percentage Continuing Use at the End of the First Year

From Trussell J, Aiken ARA. Contraceptive efficacy. In: Hatcher RA, Nelson AL, Trussell J, et al, eds. *Contraceptive Technology*. 21st ed. New York: Ayer Company Publishers; 2018.

Method	% of Women Experiencing an Unintended Pregnancy Within the First Year of Use		% of Women Continuing Use at 1 Year [‡]
	Typical Use [*]	Perfect Use [†]	
No method [§]	85	85	
Spermicides	21	16	42
Female condom [¶]	21	5	41
Withdrawal	20	4	46
Diaphragm ^{**}	17	16	57
Sponge	17	12	36
Parous women	27	20	
Nulliparous women	14	9	
Fertility awareness–based methods ^{††}	15		47
Ovulation method [‡]	23	3	

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Combined and progestin-only pills	7	0.3	67
Evra patch	7	0.3	67
NuvaRing	7	0.3	67
Depo-Provera	4	0.2	56
Intrauterine contraceptives			
ParaGard (copper T)	0.8	0.6	78
Skyla (13.5 mg LNG)	0.4	0.3	
Kyleena (19.5 mg LNG)	0.2	0.2	
Liletta (52 mg LNG)	0.1	0.1	
Mirena (52 mg LNG)	0.1	0.1	80
Nexplanon	0.1	0.1	89
Tubal occlusion	0.5	0.5	100
Vasectomy	0.15	0.1	100

This chapter presents contraceptive methods according to efficacy, with the methods that most effectively prevent pregnancy presented first followed by the less effective methods. This framework of contraceptive effectiveness follows the communication tool offered by the WHO (Fig. 13.1) but is not a one-size-fits-all formula for counseling. For example, one study found lack of side effects and affordability were key to acceptability, in addition to efficacy (Lessard, 2012). Identifying preferences is key because a patient may feel misunderstood or reluctant to seek health care if her provider emphasizes effectiveness when she places greater value on another feature such as bleeding patterns.

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FIG. 13.1

The World Health Organization's tiered approach contraception counseling tool comparing typical effectiveness of contraceptive methods. *IUD*, Intrauterine device.

(Redrawn from Association of Reproductive Health Professionals. You Decide Tool Kit: Contraceptive Efficacy Tools. Adapted from World Health Organization. *Comparing Typical Effectiveness of Contraceptive Methods*. Geneva, Switzerland: WHO; 2006.)

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awaiting contraceptive protection. When a health care provider is not reasonably certain about pregnancy, the benefits of initiating methods such as the birth control pill, implant, or injectable likely outweigh the risks. For intrauterine devices, the patient and provider should use SDM to explore the benefits of the device against the risks of an unlikely but possible pregnancy. After a quick start, initiators should be encouraged to repeat a pregnancy test if menses does not return or if pregnancy symptoms emerge. Occasionally pregnancy may occur after initiating a method using quick start. Contraceptive hormones are not teratogenic, and it is safe to continue pregnancies after a quick start or in other instances of contraceptive failure if desired.

Permanent contraception: Sterilization

Fallopian tube and vas deferens occlusion (sterilization) offer safe and highly effective contraception. In contrast to reversible or temporary methods, sterilization is permanent.

The decision to undergo sterilization should be made solely by the individual in consultation with a provider. The history of sterilization in the United States raises ethical concerns that must be considered, even today. Thousands of forced sterilization procedures occurred in the United States, disproportionately to poor women and women of color, whereas other women were denied sterilization because of provider beliefs (ACOG, 2017). **Prospective studies document that up to 20% of women younger than age 30 report regret after sterilization** (Hillis, 1999). If women wish to conceive after tubal sterilization, in vitro fertilization is likely necessary and is performed more often than tubal reconstructive surgery in these patients. Age should not be a barrier, however; younger women who are well informed and desire permanent sterilization can be offered the procedure. Providers offering sterilization should understand federal and local rules related to waiting periods, age, and disability status.

Vasectomy

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vasectomy, vas reanastomosis, a difficult and meticulous procedure, offers a success rate of approximately 50% (ASRM, 2004).

Tubal sterilization

Tubal sterilization prevents fertilization by cutting, occluding, or removing the fallopian tubes. Methods in the United States include laparoscopy anytime outside of pregnancy and postpartum sterilization concurrent with cesarean delivery or immediately after a vaginal delivery. About half of sterilization procedures occur in the postpartum period (Chan, 2010).

Tubal sterilization is highly effective, with failure rates similar to LARC methods. The Collaborative Review of Sterilization (CREST), a prospective study of more than 10,000 women who underwent transabdominal sterilization that included 14 years of follow-up, provided information related to many aspects of tubal sterilization. Findings included a 5-year cumulative failure probability of 13 per 1000 procedures, with failures sometimes occurring years later. Younger women had a higher risk of failure. Postpartum partial salpingectomy carried the lowest 10-year cumulative risk of failure (7.5 per 1000 procedures); the rate with bipolar fulguration was higher (Peterson, 1996).

The CREST study occurred before the widespread adoption of Filshie clips, transcervical sterilization, and salpingectomy techniques. Some data show a 10-year cumulative failure rate of **2 to 3 per 1000 procedures for the Filshie clip** (Shaw, 1999). Data are less clear for transcervical sterilization; Gariepy calculated a first-year failure rate of 57 in 1000 (Gariepy, 2014). Clinicians lack sufficient information on expected failure rates for salpingectomy; however, the low failure rate of partial salpingectomy may serve as a proxy. **If pregnancy occurs after tubal sterilization, location should be ascertained immediately because these pregnancies are often ectopic.**

Surgical approach

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FIG. 13.2

Modified Pomeroy technique of female sterilization.

(From Sciarra JJ. Surgical procedures for tubal sterilization. In: Sciarra JJ, Zatuchni GI, Daly MJ .
Gynecology and Obstetrics. Vol. 6. Philadelphia: Harper & Row; 1984.)

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Tubal sterilization with occlusion methods reduces the risk of ovarian cancer. Strong data suggest that many serous, endometrioid, and clear cell carcinomas begin in the fallopian tube rather than the ovary. Opportunistic salpingectomy, or the removal of both fallopian tubes in their entirety for the purposes of ovarian cancer risk reduction, appears to be a safe method of permanent sterilization with risks comparable to occlusion methods of tubal ligation. The risks and benefits of both techniques should be discussed with patients who desire permanent sterilization (ACOG, 2019).

Long-acting reversible (LARC) methods

The intrauterine device and implant (LARC methods) are highly effective and provide an option of years-long use without further clinician visits or medication refills. Unlike sterilization, **LARC methods are immediately reversible with a rapid return to fertility** . Very few medical contraindications to LARC exist. Although upfront costs can be high, these methods do not incur maintenance costs. LARC users typically report high continuation rates and user satisfaction. **The American Congress of Obstetricians and Gynecologists (ACOG) recommends that it is safe to offer nulliparous women and adolescents LARC methods** . (ACOG, 2018).

Practitioners should be mindful that women can face biased or even coercive contraceptive counseling practices and feel pressured to choose long-acting methods like LARC based on the priorities of the clinician rather than the patient (Higgins, 2016). This becomes particularly important for low-income women and women of color, who more often report lower quality family planning health care interactions than their higher-income and white counterparts. In some studies, providers were more likely to recommend IUDs to black and Hispanic women of low socioeconomic status (Dehlendorf, 2010). Conflict and coercion can arise when a patient requests LARC removal and a clinician expresses reluctance or delays or even refuses to perform a removal. Patient requests for LARC removal should be honored, regardless of clinicians' wishes.

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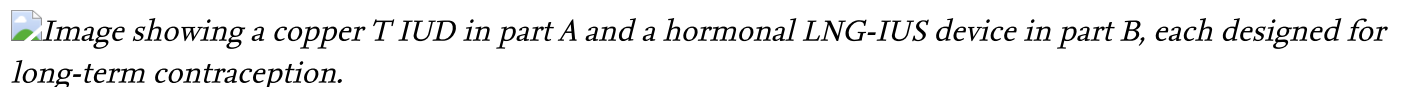
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with the annual incidence of accidental pregnancy decreasing steadily thereafter. After 12 years, the cumulative pregnancy rate of the copper T380A IUD reaches 1.7%. At 5 years of use, the pregnancy rate for the LNG-IUD is about 1.1%. These pregnancy rates are comparable to those after surgical sterilization. Contraindications to IUD insertion include pregnancy, pelvic inflammatory disease, endometritis, uterine or cervical malignancy, and a müllerian anomaly or myomas that significantly distort the uterine cavity. For additional information on medical contraindications, please see the CDC Medical Eligibility Criteria (MEC) (Curtis, 2016).

The copper T380A IUD (Paragard) (Fig. 13.3, A) is the only copper-bearing IUD currently marketed in the United States, although other copper IUDs are undergoing U.S. clinical trials. Because of the constant dissolution of copper (in amounts less than that ingested in the normal diet), copper IUDs require periodic replacement. **The copper T380A is currently approved for use in the United States for 10 years and maintains its effectiveness for at least 12 years.**

Image showing a copper T IUD in part A and a hormonal LNG-IUS device in part B, each designed for long-term contraception.

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These LNG-IUDs maintain a high level of effectiveness **for 5 years; data suggest high effectiveness continues for 7 years** (McNicholas, 2015), and both are undergoing clinical trials for further extended use. **Progestin-releasing IUDs reduce menstrual blood loss and treat heavy menstrual bleeding and related iron deficiency anemia .**

In addition, a 13.5-µg LNG-releasing IUD device with a slightly smaller body than the Liletta and the Mirena is approved for up to 3 years of use (Skyla), and another device with 19.5 mg LNG was approved for 5 years of use (Kyleena). For the purposes of this chapter, we refer to the with 20-µg releasing LNG IUD devices when discussing the LNG-IUD.

Mechanisms of action

All IUDs induce a local and sterile inflammatory response within the endometrium that incapacitates sperm thereby preventing fertilization of the ovum. This sterile inflammatory response was the only mechanism of inert historical IUDs. **Modern IUDs contain either copper or progestin that augments local effects and improves efficacy. No IUD causes abortion; all contraceptive effects are preimplantation.**

Copper markedly increases the extent of the inflammatory reaction with accumulation throughout the uterine lumen, the cervix, and probably the fallopian tubes. In addition, **copper impedes sperm transport and viability in the cervical mucus** . Because of these changes, very few, if any, sperm reach the oviducts, and the ovum usually does not become fertilized. The very small numbers of fertilizations that do occur underlie the failure rate of these devices. The **primary effect of the progestin in the LNG-IUD is to thicken cervical mucus** . Thickened mucous impedes sperm mobility and access to the upper genital tract. Additionally, **the LNG-IUD decreases tubal motility and produces a thin, inactive endometrium** . The low levels of circulating steroid sometimes inhibit ovulation. Systemic LNG levels are lower in IUD users than users of LNG-containing implants or oral pills.

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although a paracervical block may decrease it, particularly in nulliparous women. Clinicians should receive training in correct insertion technique as detailed in the product labeling

Uterine bleeding

Most women who discontinue use of the copper IUD do so for heavy or prolonged menses or intermenstrual bleeding related to increased local prostaglandin release in response to the IUD. **On average, copper IUD users experience about 1 additional day of bleeding per cycle . Heavy bleeding decreases with time and rarely leads to anemia . In contrast, LNG-IUD users experience a reduction in menstrual bleeding as early as 3 months after insertion that persists for the duration of use . After 24 months, 50% of users report amenorrhea and 25% oligomenorrhea ;** providers should discuss the possibility of eventual amenorrhea with patients before insertion because some may find this unacceptable.

Uterine perforation

Uterine perforation is a rare (**1 in 1000 insertions**) but potentially serious complication. Perforation is usually fundal and **occurs at the time of insertion** . IUDs correctly inserted entirely within the endometrial cavity do not migrate through the uterine muscle into the peritoneal cavity. With experienced providers, the risk of perforation is less. Straightening the uterine axis with a tenaculum and measuring the cavity with a uterine sound before IUD insertion reduces perforation risk. **Perforation risk is slightly higher in women currently breastfeeding.**

The clinician should rule out perforation if the user cannot feel the IUD strings and did not observe that the device was expelled. To assess missing strings, the provider can use ultrasound to locate the device. **If the device is not located with ultrasonography, a radiograph visualizing the abdomen and pelvis should be performed ;** all IUDs will be visible on a plain radiograph. IUDs outside the uterus usually can be removed by means of laparoscopy. In a study of 61,448 women with at least a year of follow-up, no IUD perforations led to serious clinical complications beyond the need for laparoscopic retrieval (Heinemann, 2015).

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If intrauterine, the device should be removed. **Pregnancies that occur with an IUD in place have a higher incidence of spontaneous abortion or infection** . After IUD removal, the complication rate becomes similar to that of a pregnancy without an IUD. In the case of an undesired pregnancy, a vacuum aspiration can be performed to remove both the pregnancy and the device.

Infection

The risk of infection after IUD insertion is small and decreases with time . The placement process, not the device itself or its threads, creates a transient risk of infection, as does any transcervical procedure. In a meta-analysis of randomized controlled trials (RCTs), routine use of prophylactic antibiotics preceding IUD insertion did not change the risk of pelvic infection (Grimes, 1999). An IUD may be placed without results from cervical screening for infection; however, if a provider has clinical suspicion of infectious endocervicitis, testing for gonorrhea and chlamydia should be performed and the IUD insertion delayed. **Providers caring for women undergoing routine insertion or care with an IUD in place should follow standard sexually transmitted infection (STI) screening guidelines.**

Positive gonorrhea or chlamydia screening tests that occur with an IUD in place can be successfully treated without removing the IUD. For patient with symptoms of pelvic inflammatory disease (PID) and an IUD, an antibiotic regimen for PID approved by the CDC should be used with close clinical follow-up. If the infection does not improve within 48 to 72 hours of treatment or if there is evidence of tubo-ovarian abscess, the device should be removed and an alternative method initiated (Hauk, 2015).

Routine cytologic testing can identify *Actinomyces* organisms during cervical screening with IUDs in place. If the woman is asymptomatic, no treatment is indicated. In the rare event of pelvic infection, treatment includes IUD removal and an appropriate course of antibiotics (usually penicillin).

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hormone rank, among the most effective methods of contraception, with an effectiveness equal or superior to that of sterilization and IUDs. In the United States one implant is available, **Nexplanon, which contains 68 mg of etonogestrel (ENG)**. Nexplanon is approved for **3 years of use**; however, preliminary studies indicate continued effectiveness for 5 years (McNicholas, 2015). The **implant inhibits ovulation and thickens cervical mucus**. After removal, serum etonogestrel levels decline to undetectable within a week. Ninety percent of users ovulate within 1 month after removal.

In clinical trials, continuation rates are high at 50% to 80% until 2 years. **Bleeding irregularities are the most common reason for discontinuation, accounting for about 60% of early removals**. As with other progestin-only methods, nearly all women experience changes in their regular bleeding pattern. Women may experience amenorrhea or infrequent, frequent, or prolonged bleeding. Unlike depo-medroxyprogesterone acetate (DMPA) or the LNG-IUS, the bleeding patterns of individual women will be unpredictable.

U.S. providers must complete a Food and Drug Administration (FDA)–mandated training program before providing Nexplanon implants. After skin infiltration with local anesthesia, the implant is inserted superficially into the subdermal tissue of the nondominant upper arm posterior to the groove between the bicep and triceps. The insertion site is closed with adhesive without the need for suture (Fig. 13.4), and the procedure usually takes less than a minute to complete. From the subdermal space, the steroid diffuses into the circulation at a constant rate. At the end of use, the implant must be removed. The removal site is infiltrated with local anesthetic and the implant removed through a 2- to 3-mm incision then closed with adhesive (see Fig. 13.4). Correct, superficial insertion enhances the ease of removal; improperly placed deep implants are more difficult to remove. If nonpalpable, an implant can be identified using imaging of the upper arm. Rare cases (1 per 100,000 insertions) of distant migration of the implant to the pulmonary artery have been reported (Reed, 2019).

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FIG. 13.4

Insertion of a contraceptive implant.

(Modified from and used with permission of Mayo Foundation for Medical Education and Research. All rights reserved.)

Hormonal contraceptives methods mechanisms of action

Systemically absorbed hormonal contraceptives primarily prevent pregnancy by suppressing gonadotropins. The progestin component suppresses the luteinizing hormone (LH) surge, which inhibits ovulation. Methods that contain estrogen prevent a rise in follicle-stimulating hormone (FSH), which inhibits follicular development through a synergistic effect with the progestin. Estrogen also stabilizes the endometrium, which prevents unscheduled bleeding. In addition,

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health conditions will decrease with coadministration of estrogen-containing contraceptives. Specific prescribing and interactions guidelines are available through the CDC MEC.

Hormonal method side effects

Some hormonal contraception users report various negative physical and psychological side effects such as headache, weight gain, mood disorders, breast changes, bloating, and sexual problems. Such effects may lead to method discontinuation. Clinicians should be attentive to an individual's symptoms and concerns, offer reassurance, and be willing to change to a different contraceptive on request. In discussing these experiences, providers and patients should also recall that negative physical feelings and symptoms are very common among women of reproductive age who do not use hormonal contraception. Research from placebo-controlled studies provides insight and suggests some users can overestimate how much their method contributes. In a 6-month double-blind RCT of 507 women, those administered oral contraceptive pills or placebo reported types and numbers of side effects that were indistinguishable and a similar percentage of women discontinued over the course of the study (Redmond, 1999). **The widespread belief in such side effects may contribute to a disinclination to continue or even to start oral contraceptives.**

Noncontraceptive health benefits of hormonal contraceptives

In addition to preventing pregnancy, hormonal contraceptives may provide additional health benefits to patients. **Studies consistently demonstrate a strong protective effect between progestin-containing contraceptives and endometrial cancer** . Compared with nonusers, women who use oral contraceptives (OCs) for at least 1 year have a decreased risk of developing endometrial cancer between ages 40 and 55. This protective effect is related to duration of use, increasing **from a 20% reduction in risk with 1 year of use to a 40% reduction with 2 years of use to about a 60% reduction with 4 years of use** . This protective effect **persists for at least 15 years** after stopping use of the OCs.

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improves during use of hormonal methods because progestins suppress endometrial proliferation, and hormonal methods that inhibit ovulation can also reduce ovulatory disorders such as Mittelschmerz and premenstrual syndrome.

Estrogen-containing methods of birth control such as OCs, the patch, or the ring may improve acne. The effect of a method on acne relates to three properties: the androgenicity of the progestin component, the reduction of free androgens as a result of an estrogen-driven increase in serum hormone-binding globulin, and the activity of 5 α -reductase, the enzyme that converts testosterone to dihydrotestosterone. In double-blind, placebo-controlled studies, low-dose OCs had a greater reduction of acne with no difference in adverse events compared with placebo pills (Redmond, 1997).

Hormonal contraceptives may be ideal treatments for gynecologic conditions such as polycystic ovary syndrome (PCOS) and pelvic pain. In the setting of PCOS, estrogen-containing methods reduce circulating androgens by suppressing gonadotropins and inducing the production of sex hormone-binding globulin. All progestin-containing methods protect the endometrium from an arrested proliferative phase that is common in PCOS and the concurrent increased risk of endometrial hyperplasia.

Injectable suspensions

DMPA (Depo-Provera) is the only injectable contraceptive available in the United States.

Worldwide, other short-term injectables composed of progestin alone or a combination of progestins and estrogens are also available.

Medroxyprogesterone acetate (MPA) is a long-acting derivative of progesterone formulated as an injectable crystalline suspension. **When used consistently, the failure rate at 1 year is 0.2%.**

Typical use **failure rates are around 6 %**, related to delayed injections. These effectiveness rates apply to women of all body weights.

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Return to fertility may be delayed because of the lag time in clearing DMPA from the circulation. The median delay to conception is 9 to 10 months after the last injection, with a wide range in resumption of ovulation, from 15 to 49 weeks from the last injection. Women who wish to become pregnant after discontinuing DMPA may experience a delay in the resumption of fertility. After this initial delay, fecundity is similar to that found after discontinuing a barrier contraceptive (Fig. 13.5). Use of DMPA delays but does not prevent return of fertility.

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FIG. 13.5

Cumulative conception rates of women who discontinued a contraceptive method to become pregnant. *IUCD*, Intrauterine contraceptive device.

(From Schwallie PC, Assenzo JR. The effect of depo-medroxyprogesterone acetate on pituitary and ovarian function, and the return of fertility following its discontinuation: a review. *Contraception*. 1974;10(2):181-202.)

Clinical side effects

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FIG. 13.6

Percentage of patients with bleeding or spotting on days 0, 1 to 7, 8 to 10, or 11 to 30 per 30-day cycle while receiving injectable depo-medroxyprogesterone acetate (DMPA), 150 mg every 3 months.

(From Schwallie PC, Assenzo JR. Contraceptive use: efficacy study utilizing medroxyprogesterone acetate administered as an intramuscular injection once every 90 days. *Fertil Steril*. 1973;24:331-339, from The American Society for Reproductive Medicine.)

Weight changes

About **one-fourth of women using DMPA gain weight, usually in the first 6 months of use**. Several longitudinal studies indicate that DMPA users gain between 1.5 (2.2 pounds) and 4 kg (8.8 pounds) in their first year of use and continue to gain weight thereafter. The mechanism of weight gain in women using DMPA is poorly understood, and some suggest that the injection be used with caution in patients who are already overweight or obese.

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DMPA health benefits

In addition to those noncontraceptive benefits that apply to all hormonal methods, DMPA can be used for women with hormonally sensitive (catamenial) epilepsy to suppress estrogen fluctuations and avoid progestin withdrawal that triggers seizures in spontaneously cycling women. In some studies, DMPA seems to have beneficial effects on sickle cell pain crises.

Oral contraceptives

Effective and easy to use, OCs became the most widely used reversible contraceptive method within a few years of their introduction to the United States in 1960. The high-dose original pill formulations caused side effects such as nausea, breast tenderness, and weight gain that often led to discontinuation. Since that time, other formulations with steadily decreasing dosages of both the estrogen and progestin components have become available. **Reductions in the dose of the contraceptive steroid ethinyl estradiol (EE) has coincided with a lower incidence of severe adverse cardiovascular effects and side effects without increasing the failure rate** . Today, the most widely used OCs combine EE with one of several synthetic progestins.

Pharmacology and formulations

The three major types of OC formulations include daily progestin-only pills (POPs), also known as minipills, fixed-dose (monophasic) combination pills, and multiphasic combination pills. The combination formulations are the most widely used and incorporate several different types of progestins. **The modifications in chemical structure of different synthetic progestins affect their biologic activity** . One should consider biologic activity, not just the amount of steroid present, to characterize the pharmacologic activity of the progestin. Monophasic products contain tablets with the same dose combination of an estrogen and progestin each day. In multiphasic

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Other dosing strategies include extended and continuous cycles. Extended cycle regimens contain 84 days of active pills followed by a 7-day HFI that results in scheduled withdrawal bleeding only four times a year. Another product provides a continuous daily regimen with 28 active pills to eliminate scheduled withdrawal bleeds. Some providers recommend extended use of generic, less expensive monophasic OCs, instructing patients to discard the placebo pills and begin a new cycle pack after 21 days for a similar effect. This strategy assumes access to early refills.

Effectiveness

OCs have a 1% failure rate with perfect use and an 8% failure rate with typical use. In a cohort study of U.S. women, 24-day OC regimens containing a progestin with a long half-life had higher contraceptive effectiveness compared with 21-day regimens (Dinger, 2011). No other difference in clinical effectiveness has been demonstrated among the various combination formulations available in the United States.

Clinicians should ensure easy access to refills. Accidental pregnancies occurring during OC use probably reflect delayed initiation of a new pill pack or missing more than one or two tablets. The pill-free interval must not extend beyond 7 days; to maintain ovulatory suppression the most important pill to take on time is the first one of each cycle. When a woman misses two or more pills in a pack, she should take emergency contraception and use backup for a week.

Physiology

The balance between estrogen and progestin influences the bleeding profile of a combination OC. Estrogen induces endometrial proliferation. Progestins oppose the mitotic action of estrogen, leading to a thinner and stable decidualized endometrium. Even though women taking combined OCs are exposed to both hormones at the same time rather than sequentially as in a

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over time, with rates of unscheduled bleeding highest in early cycles. Women who experience prolonged breakthrough bleeding during continuous OC use might benefit from discontinuing the active pills for 3 days to allow for a short period of withdrawal bleeding and then restarting. Greater gaps could result in ovulation and unintended pregnancy.

Metabolic effects

Weight gain represents a common experience of women using hormonal contraception. Weight gain is common in the U.S. population, and the extent to which hormonal methods cause additional weight gain appears limited. Many OC users do not gain weight; one 2013 study assessed weight after OC initiation in women with obesity and normal weight, weighing participants before and after 3 months of OCP use. Neither group experienced a substantial change in weight (Mayeda, 2014).

Return to fertility

After discontinuation of low-dose OCs, the suppressive effect on the hypothalamic-pituitary-ovarian axis disappears quickly. After the initial recovery, completely normal endocrine function occurs. There is little, if any, effect of duration of OC use on the time to subsequent conception. There is no risk of congenital malformations or other adverse outcomes in pregnancies among women who conceive while taking OCs or shortly thereafter.

Neoplastic risks and benefits

For a discussion of neoplastic risks and benefits, see Noncontraceptive Benefits of Hormonal Contraceptives earlier in this chapter.

Breast cancer

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Cervical cancer

The epidemiologic data regarding the risk of invasive cervical cancer and cervical intraepithelial neoplasia with OC use are conflicting. **Confounding factors could account for contrasting results in different studies . Studies indicate that the risk of both invasive and in situ cervical cancer increases with increasing duration of OC use .** These risks persist after adjusting for human papillomavirus (HPV) infection, squamous cell or adenocarcinoma, number of sexual partners, cervical cytology screening, smoking, and use of barrier contraceptives (Appleby, 2007 ; Smith, 2003).

There is no evidence that OC use alters the incidence or rate of the progression of cervical dysplasia to invasive cancer . Women with treated cervical dysplasia can use OCs, as can those with newly diagnosed dysplasia awaiting evaluation. More data are needed to assess the impact of OC use on cervical dysplasia and cancer in HPV-vaccinated women.

Liver adenoma and colorectal cancer

The development of a benign hepatocellular adenoma is an extremely rare occurrence. An increased risk of this tumor was reported in early OC studies of prolonged use of high-dose formulations but seems to be far lower for women taking lower OC doses. Women with active liver disease should not use hormonal contraception because the liver is a major site for the metabolism of synthetic steroids.

A meta-analysis of published studies of the relationship between OCs and colorectal cancer showed that OC ever use was associated with a 15% to 20% reduction in the risk of colorectal cancer (Fernandez, 2001).

Coagulation parameters

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Although progestins alone do not affect coagulation parameters, whether newer progestins increase the coagulation risk of OC use remains controversial. It has been suggested that COCs with the newer progestins increase the risk over that of the older, early-generation products.

Obesity is a modest risk factor for VTE (Fig. 13.7), and obesity with a body mass index [BMI] greater than 40 can be considered a relative contraindication to use of a combined hormonal method, especially in older OC users. Use of COCs by women older than age 35 who also smoke tobacco or electronic cigarettes is contraindicated because of the risk of myocardial infarction.

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FIG. 13.7

Relative risk (with 95% confidence intervals) of deep vein thrombosis by categories of body mass (BMI), adjusted for age and sex.

(From Pomp ER, le Cessie S, Rosendaal FR, Doggen CJ. Risk of venous thrombosis: obesity and its joint effect with oral contraceptive use and prothrombotic mutations. *Br J Haematol.* 2007;139:289-296.)

Contraindications to oral contraceptive use

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migraine headache without with aura can use COC, and if headache occurs only during the hormone-free interval, continuous dosing can help control the symptoms. Women with diabetes without cardiovascular progression can take low-dose OC formulations because these agents do not affect glucose tolerance or accelerate diabetes mellitus.

Progestin-only pills

The available POP formulation consists of tablets containing a low dose of progestin, without estrogen, taken every day without a steroid-free interval. **Because the dose of progestin in POPs is less than the ovulation inhibition dose, other progestogenic effects such as cervical mucous thickening become the primary mechanism of action**. Clinicians should counsel their patients using POPs to take pills consistently at the same time of day to ensure that progestin concentrations do not fall to less than the effective level. **Newer POPs entering the U.S. market contain progestins that inhibit ovulation consistently**. Because women using POPs may still ovulate, endogenous estradiol and progesterone produced by the ovary will affect endometrial bleeding patterns, resulting in regular menses, irregular bleeding, spotting, or amenorrhea, depending on an individual woman's response.

Contraceptive patch

The contraceptive skin patch (Fig. 13.8) contains 75 µg ethinyl estradiol and 6 mg norelgestromin (Xulane). The patch is applied to the skin each week for 3 consecutive weeks followed by a patch-free week to allow scheduled withdrawal bleeding. The patch may be applied to the buttocks, upper outer arm, lower abdomen, or upper torso, excluding the breasts. After skin application, both steroids appear in the circulation rapidly and reach a plateau within 48 hours. The primary mechanism of action, effectiveness, and metabolic and clinical effects are similar to COCs. Results from early clinical trials suggest that patch efficacy may be slightly lower in women with body weight more than 90 kg; however, even in the heaviest women the patch was 90% effective. Newer patch results show similar effectiveness across body weights.

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FIG. 13.8

The contraceptive skin patch releasing ethinyl estradiol and norelgestromin.

Contraceptive vaginal ring

Steroids pass easily through the vaginal epithelium directly into the circulation. A flexible **ring-shaped device containing 2.7 mg of ethinyl estradiol and 11.7 mg of etonogestrel** (Fig. 13.9), the contraceptive ring (NuvaRing), is placed in the vagina for 21 days and then removed for up to 7 days to allow withdrawal bleeding. After this ring-free interval the woman inserts a new ring. Ring expulsion is uncommon, and both users and their partners typically report high acceptability. Mechanisms of actions, efficacy, and side effects are similar to those of COCs. Continuous use has been studied, and results are similar to those observed with continuous COCs.

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FIG. 13.9

The contraceptive vaginal ring releasing ethinyl estradiol and etonogestrel.

Clinical trials showed no difference in ring efficacy for women in the heaviest weight category (>166 lb [75 kg]) (Westhoff, 2005), but few overweight or obese women were enrolled in those trials. A pharmacokinetic study showed therapeutic progestogen and ethinyl estradiol levels adequate to inhibit ovulation without increases in follicular activity in both women with obesity and those with normal weight (Dragoman, 2013).

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no data have assessed whether a “good fit” correlates with improved effectiveness. **The FDA-approved Caya, a single-size diaphragm that does not require fitting by a practitioner, is intended for use over the counter** (Fig. 13.10).

 *Image showing a lavender-colored, contoured Caya diaphragm with grip dots used as a reusable contraceptive device.*

FIG. 13.10

The Caya diaphragm.

(Courtesy Kessel Medintim GmbH, www.medintim.de (<http://www.medintim.de>) .)

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FIG. 13.11

The female barrier method FemCap.

The diaphragm and cervical cap are used with a spermicide and left in place for at least 6 hours after the last coital act . If repeated intercourse takes place, additional spermicide should be used vaginally. The failure rate during the first year of use for the diaphragm or cervical cap ranges from 13% to 17% among all users and may be as low as 4% to 8% with perfect use. The diaphragm and cervical cap may also reduce the risk of cervical dysplasia and cancer.

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The female condom consists of a soft, loose-fitting polyurethane sheath with two flexible rings. One ring lies at the closed end of the sheath and serves as an insertion mechanism and internal anchor for the condom inside the vagina. The outer ring forms the external edge of the device and remains outside the vagina after insertion, thus providing protection to the introitus and the base of the penis during intercourse (Fig. 13.12). The female condom is prelubricated and intended for single use. Like male condoms, the device is available over the counter, does not require fitting, and protects against sexually transmitted infections. It can be inserted before the onset of sexual activity and left in place after ejaculation has occurred. The typical use failure rate at 1 year is estimated to be 21% (Hatcher, 2011).

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FIG. 13.12

A female condom.

(From Beksinska M, Smit J, Joanis C, et al. Female condom technology: new products and regulatory issues. *Contraception*. 2011;83(4):316-321.)

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Fertility awareness

Many couples use abstinence from sexual intercourse or a barrier method during the “fertile window” of the menstrual cycle (the 5 days preceding ovulation or the day of ovulation) to prevent pregnancy. Several techniques are used to estimate the timing of ovulation.

The oldest of these is the calendar rhythm method. With this method, the period of abstinence is determined by calculating the length of the individual woman’s previous menstrual cycle and using three assumptions: (1) The human ovum can be fertilized for only about 24 hours after ovulation, (2) sperm can fertilize for 3 to 5 days after coitus, and (3) ovulation usually occurs 12 to 16 days before the onset of menses. The woman therefore establishes her fertile period by subtracting 18 days from the length of her previous shortest cycle and 11 days from her previous longest cycle and abstains from coitus during this time. This method assumes a normal and predictable menstrual cycle.

Other fertility awareness methods rely on cyclic physiologic changes. Increasing levels of progesterone occurring after ovulation cause a detectible rise in **daily basal body temperature (BBT)**. Using the BBT method, the woman must abstain from intercourse from the cessation of menses until the third consecutive day of elevated basal temperature, or when she is postovulatory. The cervical mucus method requires that the woman recognize and interpret the presence and consistency of cervical mucus. Increasing estradiol levels increase the production of cervical mucus. Intercourse can occur after menses ends until the first day that copious, slippery mucus is observed to be present and again 4 days after the last day when the characteristic mucus was present.

The calendar, temperature, and cervical mucus methods can be used separately or in combination with one another—the symptothermal method. Overall typical failure rates are around 24%. Women with irregular cycles, those who are older than 35, and those who are immediately postpartum should not use fertility awareness methods that rely on regular cycles and predictable ovulation. Women using these methods should also have control over when intercourse occurs.

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~~caution because most are inaccurate and may even contain misleading information. Patients~~
should also be cautioned about uncertain privacy protections for their personal information.

Coitus-related methods

Spermicides

Spermicides consist of an active agent and a carrier. The active agent is a surfactant that immobilizes or kills sperm on contact by destroying the sperm cell membrane. The carriers include gels, foams, creams, tablets, films, and suppositories. Spermicides must be placed into the vagina before each coital act, often in combination with a barrier contraceptive to increase effectiveness. The contraceptive sponge, a cylindric piece of soft polyurethane impregnated with 1 mg of **nonoxynol-9 spermicide**, must be inserted into the vagina before intercourse and is effective for 24 hours. **The failure rate ranges from 15% to 25%.**

Coitus interruptus (withdrawal)

Removal of the penis from the vagina before ejaculation to prevent pregnancy is an ancient male-controlled method of contraception without contraindications, devices, or cost. Withdrawal can fail because of sperm present in some preejaculate, the fluid produced by the penis during sexual excitement and before ejaculation. More commonly, the method fails if withdrawal is not performed in a timely fashion. Correct and consistent use with every act of intercourse should be stressed, not only during suspected fertile times. A major drawback of the method is the lack of any protection against sexually transmitted infections. **Failure rates range from 4% with perfect use to 22% with typical use.**

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The most well-known and commonly used EC is a 1.5-mg levonorgestrel single-dose pill taken orally within 72 hours of unprotected sex. This method of EC is available over the counter and works by delaying or inhibiting ovulation. Because this effect is preovulatory, its effectiveness depends on the cycle day it is taken. If taken after ovulation has occurred, this method is no longer effective.

Another orally available EC is a 30-mg single dose of the selective progesterone receptor modulator ulipristal acetate (UPA, ella). Ovulation is delayed for 5 days in women who take ulipristal acetate. Women can take it closer to the time of ovulation than other methods and still successfully delay ovulation, even after the LH surge has occurred. Thus UPA is approved for use up to 120 hours from the time of unprotected intercourse and is more effective at preventing pregnancy than the levonorgestrel-only method (Glasier, 2010). Levonorgestrel pills are less effective as EC in women with obesity, but UPA effectiveness is maintained in these women (Praditpan, 2017).

The copper IUD is the most effective form of EC. Insertion up to 5 days after unprotected intercourse is 99% effective at preventing pregnancy. When appropriate, another benefit is the potential fulfillment of a woman's long-term contraceptive needs. Studies evaluating the LNG-IUD placement for use as EC are under way, but there is insufficient evidence to recommend its use for this purpose at this time.

Other contraceptive concerns

Postpartum

On average, resumption of ovulation with return to fertility occurs 45 days postpartum in women who are not breastfeeding. Antenatal care visits provide an ideal time to review plans and contraceptive methods. In addition to usual medical considerations, the postpartum time requires special attention to the risk of thrombosis and the impact on breastfeeding. **The risk of**

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Breastfeeding

Women who breastfeed exclusively—that is, with minimal or no nonmilk supplements—can expect contraceptive protection as a result of ovulatory suppression and resulting lactational amenorrhea for 6 months postpartum. Such exclusive breastfeeding practices remain uncommon in the United States. Providers should address contraceptive options with all women, regardless of breastfeeding intentions. Choosing the right birth control method while breastfeeding can be challenging for women who have concerns related to the impact of hormonal methods. Because withdrawal from progesterone after delivery initiates lactogenesis, use of a hormonal method of birth control while breastfeeding could theoretically affect milk production. Women and clinicians can be generally reassured; although data are limited, the LNG-IUD, the progestin-only implant, and progestin-only pills appear to have little to no impact on breastfeeding or infant growth and development (Curtis, 2016). The DMPA injection similarly appears to have little effect; however, evidence from animal studies have led the WHO to recommend delaying DMPA until 6 weeks postpartum in breastfeeding women. Clinical trials are ongoing to better answer this question.

Little high-quality research has addressed the impact of combined hormonal methods, and theoretically the effects of estrogens could reduce milk supply. Studies examining the impact of estrogen-containing contraception (pills, patch, vaginal ring) on breastfeeding have similarly inconsistent results. Some show no affect, whereas others show a reduction in breastfeeding duration and breast milk volume. These effects are no longer present once breastfeeding has been established or after 6 weeks postpartum (Bahamondes, 2013). In addition, in the first 6 weeks postpartum, estrogen-containing methods should be avoided because of the increased VTE risk. Therefore the CDC recommends a delay in estrogen-containing methods until 6 weeks postpartum in both breastfeeding and nonbreastfeeding women. In contrast, the WHO advises a delay in estrogen-containing methods until 6 months postpartum in breastfeeding women, particularly in low-resource settings, given the importance of breastfeeding on infant health. A

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abortion by age 20, 1 in 5 by age 30, and 1 in 4 by age 45. (Jones, 2017). Impressive gains in contraceptive safety, access, and use have contributed to decreasing abortion rates in the U.S. since a peak in the 1980s (Fig. 13.13). In 2017 about 862,320 abortions were performed, representing a rate of 13.5 per 1000 women, the lowest rate ever observed in the United States (Jones, 2019). Despite decreases, however, U.S. abortion rates remain higher than other Western countries.

 *Graph showing U.S. abortion rates peaking in the 1980s and steadily declining to a low of 16.9 per 1,000 women by 2011.*

FIG. 13.13

Trends in abortion. The U.S. abortion rate reached a historic low in 2017.

(From Guttmacher Institute. Induced Abortion in the United States. 2019. Available at <https://www.guttmacher.org/fact-sheet/induced-abortion-united-states> (https://www.guttmacher.org/fact-sheet/induced-abortion-united-states) .) Accessed on September 18, 2019

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SINCE THE LANDMARK *ROE V. WADE* decision that legalized abortion across the country in 1973, many states have steadily enacted laws restricting access. These state laws, which number in the hundreds, mandate waiting periods before an abortion, gestational age limits, restrictions on use of public or private insurance, and parental consent requirements, among others. Abortion care is more restricted than at any time since legalization in 1973, though a few states have increased access. Some clinics and providers who work in abortion care in the United States have experienced harassment and even overt violence.

Methods of abortion

Aspiration

Curettage by vacuum aspiration is the predominant abortion technique in the first trimester. Early pregnancy aspiration can be completed with a flexible or rigid plastic cannula connected to a syringe. This technique, called **manual vacuum aspiration (MVA)**, accomplishes uterine evacuation **up to a gestational age of 10 weeks**. Dilation of the cervix is usually required before aspiration. After 10 weeks, an **electric vacuum aspiration (EVA) system** can offer more efficient evacuation with continuous suction. In early gestations, randomized trials shown similar patient satisfaction with MVA and EVA.

After any vacuum aspiration, good practice includes careful tissue inspection to check for complete removal of the gestational sac, placenta, and fetal tissues (visible on inspection after about 9 weeks). If products of conception cannot be visualized, a sonogram can identify retained tissue. In the case of very early pregnancy, it can be difficult to identify a gestational sac. Serial human chorionic gonadotropin (HCG) evaluation or a repeat ultrasound in 1 week can ensure that pregnancy termination is complete. The MVA technique provides a low-cost and highly efficient technique to manage miscarriage in an office setting, including incomplete or missed abortion.

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commonly used dilators are small tents of latex and sterilized seaweed, *Sargassum japonica*. By gently opening the cervix over several hours, the osmotic dilators reduce the risk of uterine perforation and cervical injury. **Another strategy for cervical preparation is the administration of misoprostol, a prostaglandin and E1 analog, typically given in a dosage of 400 or 600 mg vaginally or buccally a few hours before vacuum aspiration or D&E.** Misoprostol softens the cervix to facilitate mechanical dilation. Mifepristone can be used in some cases to augment dilation protocols before D&E. Surgeons use a combination of suction and uterine forceps to remove fetal and decidual tissue. In some cases, intraoperative ultrasound guidance may assist the surgeon during D&E. Inspection to identify expected fetal tissues and the placenta confirms a complete procedure.

Medication abortion

First-trimester medication abortion

In September 2000 the U.S. FDA approved the combination of mifepristone and misoprostol as a safe and effective alternative to surgical abortion. These medications are used by a growing proportion of women undergoing early abortions. In 2017 medication abortion accounted for 39% of all nonhospital abortions and for about 60% of abortions before 10 weeks' gestation (Jones, 2019).

Mifepristone (Mifeprex), also known as RU-486, is a competitive inhibitor of the endometrial progesterone receptor. Mifepristone combined with the administration of the prostaglandin misoprostol 24 to 72 hours later results in a complete abortion more than 95% of the time. Either drug is much less effective when administered alone as a single dose. The most commonly used evidence-based regimen combines 200 mg oral mifepristone with 800 µg buccal or vaginal misoprostol. If mifepristone is not available, repeat doses of misoprostol provide another approach.

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misoprostol. When available, adding mifepristone the day before misoprostol is preferred because mifepristone substantially shortens the induction-to-delivery interval. Professional organizations recommend 200 mg mifepristone followed by 800 µg misoprostol administered vaginally 24 to 48 hours later, then 400 µg misoprostol administered vaginally or sublingually every 3 hours up to 5 doses. If mifepristone is not available, guidelines recommend a vaginal loading dose of 600 to 800 µg misoprostol followed by 400 µg misoprostol vaginally or sublingually every 3 hours up to 5 doses (ACOG, 2013). This regimen appears to be safe in women with a prior cesarean delivery because uterine rupture is very rare. Some advantages of a medical induction of labor over D&E include the avoidance of surgery and the chance to view or hold the fetus if desired. If no provider skilled in D&E is available, labor induction may be the only safe option. Access to adequate pain control throughout the process is a key consideration. Disadvantages include a required inpatient hospitalization, increased costs compared with outpatient D&E, an unknown and possibly prolonged induction interval (can exceed 24 hours), and a higher risk of retained placenta necessitating D&C in about 15% of cases.

Approaches to pain control

For surgical abortion, providers should discuss available options to weigh associated risks and benefits. Options can range from local to general anesthesia. For early-term office-based procedures, many women do well with a paracervical block, usually using lidocaine, and an NSAID; some providers add anxiolytics. If available, moderate or deep sedation can also be offered. Some degree of sedation is usually required for procedures in the midtrimester. Administering sedation requires attention to maintaining safe respiration.

Complications

Modern, legal abortion in the United States is very safe (National Academies of Science, 2018). **Serious complications are rare, and the overall mortality is less than 1 per 100,000 procedures .** The U.S. maternal mortality rate, by comparison, was 17.2 per 100,000 births from 2011 to 2015 (Petersen, 2019). Two important determinants of complications are the gestational age and

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indicates abortion does not increase a women's risk of infertility, abnormal placentation, breast cancer, depression, or anxiety (National Academies of Science, 2018).

Suggested readings for this chapter can be found on ExpertConsult.com (<http://www.ExpertConsult.com>)

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